

ADITHI JAYARAMAN

Robotics Engineer · BSc Robotics · Falmouth University

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📌 Open for sponsorship

Professional Summary

Final-year Robotics Engineering student at Falmouth University with hands-on expertise in ROS2 systems, reinforcement learning, computer vision, and hardware-software integration. I design and develop intelligent robotic platforms, from gesture-controlled arms to RL-driven companion robots that adapt through human interaction. My dissertation, awarded a first class grade, benchmarks Q-Learning and PPO in MuJoCo to navigate a UR7e arm through narrow mazes, comparing autonomous RL performance against human control to advance sim-to-real manipulation. I am deeply committed to experimenting, optimising, and solving challenges across control loops, sensor fusion, and robot autonomy, eager to bring that energy to a forward-thinking Research and Development or robotics systems team.

Technical Skills

Programming & Frameworks

› Python › C++ › C# › MATLAB › R
› ROS2 › Gazebo › Webots
› OpenCV › MediaPipe
› Reinforcement Learning › Path Planning
› Git › Agile/Scrum › L^AT_EX

Hardware & Design

› Fusion 360 (CAD) › KiCAD › EasyEDA
› Raspberry Pi › ODrive Actuators
› Sensor Integration › Embedded Systems
› Mechatronic System Design
› Iterative Prototyping & Testing

Professional Experience

Development & Prototyping Intern → Part-Time Engineer

Jul 2025 – Dec 2025

Ungraded · Falmouth, UK

- Completed a 90-hour summer internship building a hardware–software prototype to automate part of Ungraded’s internal workflow, working on a confidential imaging/automation system under NDA.
- Extended into a part-time role alongside final-year studies, contributing to sensor integration, iterative design, and testing in a fast-paced R&D environment.
- Maintained technical documentation and version control (Git) throughout; collaborated directly with the core engineering team on requirement refinement and troubleshooting.

Senior Ambassador

Jun 2024 – Present

Falmouth University · Falmouth, UK

- Mentored a team of student ambassadors; led planning and execution of open days and campus events, demonstrating leadership and coordination in high-visibility environments.
- Communicated complex engineering programmes to diverse audiences — including prospective students, parents, and industry partners with clarity and adaptability.

Robotics Course Representative

2023 – Present

Falmouth University · Falmouth, UK

- Elected by peers to represent the Robotics cohort, acting as the primary liaison between students and faculty to raise concerns, improve the course experience, and drive meaningful change.
- Attended departmental meetings, gathered student feedback, and advocated for curriculum improvements directly relevant to industry and research.

General Secretary — Robotics Society

2023 – Present

Falmouth University · Falmouth, UK

- Managed day-to-day operations of the Robotics Society, coordinating meetings, communications, and events that brought together engineering students across disciplines.
- Helped grow the Society’s presence on campus, organising technical workshops, project showcases, and networking opportunities for members.

- Led cultural engagement for Asian students, planning and executing events celebrating Asian heritage while building cross-institutional partnerships to expand the Society's reach and impact.
- Developed strong stakeholder management, advocacy, and community-building skills — particularly relevant in Singapore's diverse, multicultural professional environment.

Key Projects

SOLARIS — Human-Interactive Robotic Arm

Python · MediaPipe · OpenCV · Fusion 360

- Designed and built a 3-DOF robotic arm controlled via real-time hand-gesture tracking using MediaPipe and OpenCV, converting live pose data into low-latency motor commands.
- Engineered the full mechanical structure in Fusion 360, ensuring structural integrity, clean hardware–software integration, and repeatability across experiments.

Autonomous Robotic Arm Navigation (Dissertation) MuJoCo · ROS2 · Reinforcement Learning · Python · UR7e

- Implemented and benchmarked Q-Learning and PPO in MuJoCo to solve narrow maze navigation tasks, awarded a first class grade.
- Validated using a custom end effector on a UR7e arm, currently working on sim-to-real transfer of the RL policy.

Companion / Assistive Robot for Elderly Care

RL · Embedded Sensing · Behaviour Policy Design

- Designed a soft, personalised companion robot that adapts movement, sound, and haptic responses over time via reinforcement learning — optimising for user comfort and engagement.
- Integrated embedded touch and proximity sensors with a learned behaviour policy that responds to individual interaction patterns.

Warthog-Inspired Mobile Robot Platform

Raspberry Pi · ODrive · ROS2 · Teleoperation · Digital Prototype

- Built a ruggedised animatronic field robot for research in Africa, integrating high-torque ODrive actuators, Raspberry Pi control, and radio remote operation for robust teleoperation.
- Designed a powered scissor-lift for payload transport; implemented modular architecture ready for semi-autonomous operation and advanced sensing upgrades.

Education

BSc Robotics

Falmouth University · Falmouth, UK

Sep 2023 – July 2026 (Expected)

- **Dissertation:** Autonomous Robotic Arm Navigation — Reinforcement Learning vs. Human Control
- **Key modules:** Data Fundamentals, Computational Mathematics, Digital Prototyping, Robotics and Cybernetics, Artificial Intelligence and Machine Learning, Algorithms and Optimisation, Robot Design, Human-Robot Interaction

IB Diploma Programme

The Indian Public School · Chennai, India

Jan 2021 – Jan 2022

Certifications & Continuous Learning

- Google Project Management Certificate (Coursera, 2024)
- Machine Learning Specialisation (Coursera, 2024)
- Top 10 Skills for Robotics Engineers (LinkedIn Learning)
- The Construct — ROS2 (In Progress)
- Project Initiation & Execution (Google/Coursera)
- Bharatanatyam Arangetram Certificate

Available for Employment Pass (EP) sponsorship · Graduating July 2026 · Open to full-time and internship opportunities